Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7	*(a)*	(i)		X-rays penetrate matter **or** muscle **or** tissue (1) Denser matter not penetrated as much **or** absorbed by bone (1) X-rays affect photo film / photo cells (1)	3			3		
		(ii)	I	Rearrange to get $v = \sqrt{\frac{2eV}{m}}$ (1) v = 7.95 [or 8.0] $\times 10^7$ [m s^{-1}] (1)		2		2	2	
			II	$P = 18 \times 10^3 \times 12 \times 10^{-3} = 216$ [W] (1) X-ray power = $0.005 \times 216 = 1.08$ [W] (1)		2		2	2	
	(b)	(i)		$I = \frac{I_0}{2} \rightarrow \frac{I_0}{2} = I_0 e^{-\mu x} \rightarrow \frac{1}{2} = e^{-\mu x}$ **or** $\frac{1}{2} = e^{-\mu x}$ (1) Logs taken of both sides and convincing algebra (1)		2		2	2	
		(ii)		μ = 0.257 **or** 0.26 (1) Correct use of logs **or** $\ln 0.7 = \mu x$ (1) x = 1.39 **or** 1.4 [cm] (1)		3		3	3	
	(c)			Ultrasound B-scan would be very effective [only] form of imaging to give moving images **or** can monitor blood flow (1) Fluoroscopy also very effective with contrast medium can give a series of rapid images over time / view in real time (1) MRI (would image soft tissue) no moving images (1) X-rays poor at imaging soft tissue (1) CT scans poor at soft tissue imaging [high radiation dose] (1)			5	5		
	(d)		(i)	Absorbed dose is the energy absorbed per kg of tissue (1) Equivalent dose is the absorbed dose multiplied by a weighting factor due to the type of radiation used (1)	2			2		
			(ii)	$4 \times 0.12 = 0.48$ mSv **unit mark**	1			1	1	
				Question 7 total	**6**	**9**	**5**	**20**	**10**	**0**